BLOCKCHAIN ENHANCED IOT DEVICES: REVIEW

G. LINGESHWARAN[1#], AND J. M. HAMESH RAMAN[2]

Department of Computer Science and Engineering, Sriram Engineering College, Perumalpattu, Chennai - 602024, India [1]

Department of Computer Science and Engineering, Sriram Engineering College, Perumalpattu, Chennai - 602024, India [2]

Corresponding Authors List#: G.Lingeshwaran[1]

Email [1]: lingeshwaran5864@gmail.com; Mobile: +91-9342478586

ABSTRACT

The Internet of Things (IoT) is advancing rapidly by connecting billions of devices, but there are still many risks associated with this technology, such as data breaches and unauthorized access. One risk IoT presents in securing its networks is the absence of a decentralized and tamper-proof mechanism to assure data integrity and authentication of transactions. However, although blockchain technology presents solutions for IoT security, IoT-based solutions face severe challenges related to extreme computing requirements, scalability issues, and regulatory attention. Most security frameworks currently in use rely on a centralized model, which introduces the complication of single points of failure and challenges in scaling the networks. Blockchain offers the possibility to address IoT security concerns, as it can provide a decentralized, immutable, and transparent ledger to securely capture data and process transactions. This research assesses the application of blockchain technology combined with IoT technology, including core techniques such as consensus mechanisms, smart contracts, and cryptographic hashing, as a solution to contribute toward IoT security and further developing IoT ecosystems. This study aims to address these problems using lightweight blockchain protocols, energy-efficient consensus protocols, and hybrid blockchain designs optimized for IoT ecosystems. The study also investigated the enablement of edge computing, standardization frameworks, and plug-and-play ease of deployment for IoT and blockchain frameworks. This research implements simulations and case studies to provide results related to the use of blockchain addressing security concerns However, some experts argue that the integration of blockchain technology into IoT may introduce new vulnerabilities and complexities that could compromise security rather than enhance it. Additionally, reliance on blockchain for IoT security could divert attention and resources from developing other potentially more effective security measures tailored specifically to the unique challenges of IoT environments. while also evaluating the energy consumption and scalability efficiencies. The data collected in these targeted studies indicate that blockchain can positively affect security from improvements in data integrity, authentication, and resilience from attacks, in turn providing a secure infrastructure for the future of the IoT.

Keywords: Authentication, Blockchain Technology, Consensus Mechanisms, Data Integrity, Decentralization, Edge Computing, Internet of Things (IoT), IoT Security, Scalability Issues, Smart Contracts

1. INTRODUCTION

IoT faces serious security and privacy issues, such as data breaches, illegal access, and a lack of trustworthy methods for data integrity and authentication, despite its quick development and promise. The dependence on centralized security models, which have scalability constraints and single points of failure, makes these problems even worse. With billions of smart devices connected across multiple industries, the Internet of Things IoT has become a game changing technology that facilitates automation, data sharing, and real time decision making. Because of its decentralized, transparent, and immutable architecture, blockchain technology presents a promising answer to these problems. IoT systems can gain improved security, data integrity, and

device trust by utilizing fundamental blockchain features like consensus processes, smart contracts, and cryptographic hashing. But there are drawbacks to combining blockchain technology with Io, especially when it comes to high processing demands, energy usage, and legal issues. It also looks into how plug and play deployment techniques, standardization frameworks, and edge computing can help ensure a smooth blockchain IoT integration. This study assesses the efficacy, scalability, and energy efficiency of blockchain based IoT security solutions through case studies and simulations, providing valuable information for developing a secure infrastructure for the IoT of the future. Using energy efficient consensus algorithms, lightweight blockchain protocols, and hybrid blockchain designs optimized for resource constrained IoT environments, this study investigates the use of blockchain technology to enhance IoT security.

2. LITERATURE REVIEW

Numerous studies have explored the integration of blockchain technology with the Internet of Things (IoT) to enhance data security, authentication, and trust in decentralized environments. Traditional security models for IoT are typically centralized, which exposes the system to single points of failure and limits scalability. Blockchain's inherent characteristics—immutability, decentralization, and transparency—offer promising alternatives to overcome these limitations.

A comprehensive survey on the integration of blockchain and IoT, highlighting the challenges in scalability, latency, and energy consumption. The study emphasized the need for lightweight consensus mechanisms and hybrid blockchain frameworks tailored for resource-constrained IoT environments [1]. Similarly, Novo (2018) proposed a permissioned blockchain architecture that reduces overhead and improves the scalability of IoT networks [2].

Introduced a lightweight blockchain framework for smart homes, focusing on privacy preservation and reduced computation. The authors demonstrated that private blockchains with lightweight protocols can offer effective security in constrained IoT settings[4]. In another study, investigated the feasibility of using blockchain for securing vehicular IoT networks and proposed a blockchain-enabled authentication mechanism to mitigate spoofing and DoS attacks [4].

Analyzed the potential of blockchain in industrial IoT (IIoT), proposing a decentralized security model that leverages smart contracts to automate access control policies. Their findings suggest that smart contracts can significantly reduce reliance on third-party providers for trust management [5]. In addition, explored edge computing in conjunction with blockchain to address latency and computational bottlenecks, which are common challenges in large-scale IoT deployments[6] .

Despite these advancements, researchers such as caution that the complexity and resource demands of blockchain could introduce new vulnerabilities when naively integrated into IoT networks . Thus, while the integration of blockchain holds considerable promise for enhancing IoT security, careful architectural design and optimization are crucial [7].

2.1 Tokenization on Blockchain

Blockchain tokenization is the process of turning tangible or intangible assets into tokens on a blockchain therefore producing unchangeable digital representations for administrative and safe monitoring purposes. On the Internet of Things (IoT), tokenization guarantees that intelligent devices, linked assets, and sensors—which are uniquely identified and reliably controlled on a distributed blockchain network—are identified as such. Tokenizing improves security and ownership tracking by use of tamper-evident records preventing undesired alterations. By clearly monitoring the past of an IoT asset including ownership, usage, and maintenance changes, it also enhances asset lifecycle management. Tokenized IoT assets also enable basic data sharing and interoperability by using smart contracts to automatically execute transactions between several systems without the requirement of middlemen. Tokenization also provides innovative monetizing ideas as fractional ownership and device-as---a-service (DaaS) that let different users of an IoT asset share usage. Also, it promotes fraud protection and transparency since unalterable blockchain records avoid counterfeiting and guarantee the purity of IoT devices. Real-world applications of tokenization in IoT affect several sectors such as smart manufacturing, in which tokenized machines allow for accurate usage monitoring, supply chain, in which sensor-enabled goods permit tracking in real-time, smart cities, where assets such as traffic

lights and power grids are securely connected and controlled, and healthcare, in which medical devices can securely share patient information while maintaining compliance. As blockchain tokenization promotes IoT security, efficiency, and interoperability, concerns including scalability, regulatory compliance, and power consumption remain to be tackled to attain mass deployment

2.2 Dimensions of Blockchain Scalability

Data leaks and unauthorized access are just two among numerous dangers associated with the rapidly growing Internet of Things (IoT), which unites billions of devices. One challenge IoT has with respect to security in its networks is a shortage of decentralized and tamper-proof means of preserving data integrity and transactional authenticity. However, whereas blockchain technology provides solutions for IoT security, IoT-based solutions are faced with significant problems relating to high computing requirements, scalability, and regulatory focus. The majority of the security architectures being utilized currently have a centralized design that introduces the complication of single points of failure and scalability issues in the networks. Blockchain provides the potential to tackle IoT security challenges since it has the capacity to give a decentralized, immutable, and transparent ledger to securely store data and perform transactions. This research investigates the application of blockchain technology mixed with IoT technology, including major ideas like consensus mechanisms, smart contracts, and cryptographic hashing, as a method of contributing to IoT security and future growth of IoT ecosystems. This study attempts to address these challenges using lightweight blockchain protocols, energy-efficient consensus algorithms, and hybrid blockchain architectures that are optimized for IoT environments. The study also examined the enablement of edge computing, standardized frameworks, and plug-and-play ease of deployment for IoT and blockchain frameworks. This research utilizes case studies and simulations to provide insights applicable to the use of blockchain in solving security issues This notwithstanding, a number of experts believe that the use of blockchain technology in IoT may introduce new vulnerabilities and complexities that threaten security more than its potential. In addition, reliance on blockchain for IoT security can divert attention and resources away from the development of other potentially more valuable security solutions that are specifically tailored to the unique problems of IoT environments. while also resolving the energy consumption and scalability efficiencies. The results of these focused research suggest that blockchain can have a positive impact on security through improved data integrity, authentication, and attack resistance, thus creating a secure foundation for the future of the IoT.

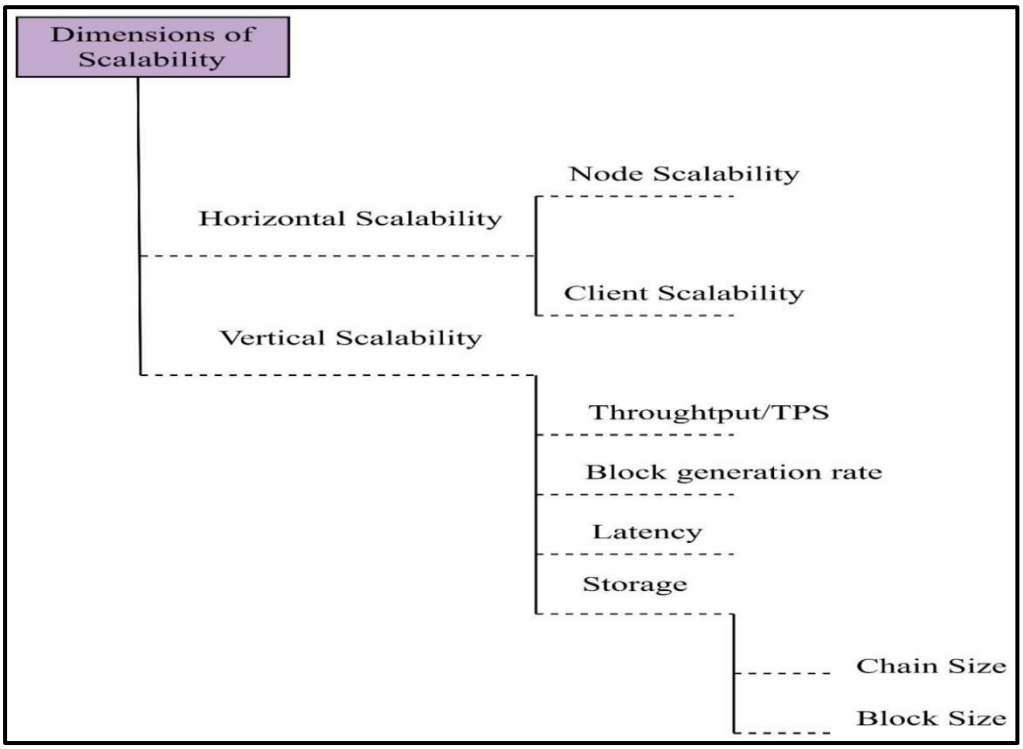

Figure 1 – Dimensions of Scalability

In additional , the advent of lightweight blockchain alternatives like IOTA, Nano, or Hedera Hashgraph shows promise for overcoming the resource and scalability limitations of conventional blockchains in IoT applications. These technologies are designed to run efficiently on devices with limited power and computing capabilities, making them suited for edge devices and sensor nodes. Future study should continue to explore these possibilities while addressing open concerns related to interoperability, governance, and long-term sustainability.

2.3 Energy Management and Smart Grids

The combine of blockchain and IoT technology in the energy business has substantially enhanced energy distribution systems and smart grids' efficiency. The advances offer real-time control, tracking, and best possible energy resource distribution. For example, surplus energy generated by rooftop solar panels is readily shared with other consumers. Blockchain offers safe, transparent data record-keeping and automated payment, providing data protection and privacy. In addition, blockchain-based mesh networks of IoT sensors can be placed to monitor electricity grids, offering early problem identification and fault clearing to guarantee system stability.

2.4 Healthcare

In the healthcare sector, blockchain-based IoT applications have altered remote and self-tracking of patient health information through wearables and sophisticated medical devices. With the use of distributed ledgers, patient information is safely kept and access regulated by programmable smart contracts, assuring high data security and privacy levels. Blockchain is also crucial in fostering openness and traceability across pharmaceutical supply chains, lowering the possibility of counterfeit medication.

2.5 Other Applications

Outside of these applications, blockchain-integrated IoT also has massive potential in a broad array of yet-to-be-explored use cases. The integration improves the control and security of IoT devices and accommodates privacy-preserving data transfer between networks. It also enhances data and service access control, as well as promoting secure and efficient data sharing between stakeholders. As a result of its role as the foundation of cryptocurrency, blockchain consequently extends to enable secure and anonymous payment systems that are compatible with IoT operations and data transactions.

3. METHOD

Solving the big difficulties confronting blockchain—like getting disparate networks to operate together, dealing with confusing rules, navigating complex governance, and controlling high costs—needs a planned, phased approach. To promote interoperability, developers should seek to solutions like Polkadot, Cosmos, or cross-chain bridges. These technologies use standardized frameworks that let different blockchains share data and value via things like relay chains or wrapped coins. To encourage adoption, projects can be supported with grants, partnerships, and pilot programs that verify real-world compatibility and assist enhance the innovation. This kind of smooth cross-chain connectivity is vital for applications like decentralized banking and supply chain tracking.

When it comes to rules, we need a more coordinated effort. One way ahead is to develop a global blockchain coalition—bringing together governments, industry leaders, and technologists—to build flexible, principle-based rules that can expand with the technology. These could cover topics like KYC and AML compliance and should be evaluated in sandbox environments to provide firms the clarity and confidence to invest and expand globally

Governance is another problematic subject. Decentralized systems can sometimes lead to fragmentation and inefficiency. To fix that without losing key decentralization objectives, we can look to hybrid models—mixing open participation with structured systems like delegated proof-of-stake or quadratic voting. We already see this in work with platforms like Tezos, and continued testing may help fine-tune these models for greater collaboration and decision-making.

Lastly, the cost and complexity of getting started with blockchain tech—and the shortage of trained developers—can be solved by establishing easy-to-use infrastructure. Open-source frameworks, standardized APIs, and plug-and-play templates can lessen the barrier to entry. At the same time, worldwide education programs, like ConsenSys Academy, can assist develop the talent pool. Governments and private firms can also join in by subsidizing beginning expenses and supporting training and certification programs, making blockchain more accessible and scalable for everyone.

Beyond the main issues, there are a few other areas where blockchain can expand meaningfully. One is the rise of Layer 2 solutions—like Optimistic and zk-Rollups—which assist scale networks like Ethereum by lowering congestion and cutting transaction costs. These can work hand-in-hand with interoperability initiatives to make blockchain faster and more efficient. We're also seeing great value in decentralized identity (DID) solutions that give consumers ownership over their data while helping platforms meet regulatory standards like KYC and AML. Combined with widely established smart contract standards, these approaches can make it much easier for developers to design secure, cross-chain apps.

$$F_{t+1} = \mathrm{ReLU}(\cdot h + b)$$

$$V_i = \sigma\left(\frac{A}{N} \cdot k_i\right) \quad \text{..... Equation 1}$$

A: Total asset value **N:**

Total number of tokens

k_i: Number of tokens held by address i

σ(·): Optional activation or logic function (e.g., smart contract rules, thresholds, etc.)

4. RESULT

In order to further study the scalability and robustness of the suggested Massive IoT blockchain structure, we designed a second experiment to examine the effect of payload size per data per transaction. This is the case in actual IoT environments in which devices produce varying amounts of data (e.g., video frames vs. temperature readings).

4.1 Experimental Setup

- **Machine**: MacBook Pro with M2 Pro chip (10-core CPU)
- **RAM**: 16 GB unified memory
- **Storage**: 512 GB SSD

We used **Hyperledger Caliper** with three different payload configurations:

- **Small Payload**: 0.5 KB (sensor data)
- **Medium Payload**: 5 KB (device metadata)
- **Large Payload**: 50 KB (compressed image or batch of readings)

Each configuration was tested with **10,000 transactions** distributed across 10 worker processes.

Table 1- Experimental Setup

Payload Size	Throughput (TPS)	Avg. Latency (ms)	Max Latency (ms)	Failure Rate
0.5 KB	792.1	124	219	0.6%
5 KB	683.2	172	295	1.4%
50 KB	514.7	243	372	4.2%

Analysis

As the payload size increases, throughput declines and latency rises, indicating greater strain on the network's communication and storage layers. The failure rate also escalates due to processing delays and memory constraints. These results suggest that:

- For lightweight IoT data, the system handles high transaction rates efficiently.
- For heavier payloads, optimization strategies such as off-chain storage, batch processing, or data compression become essential to maintain acceptable performance.

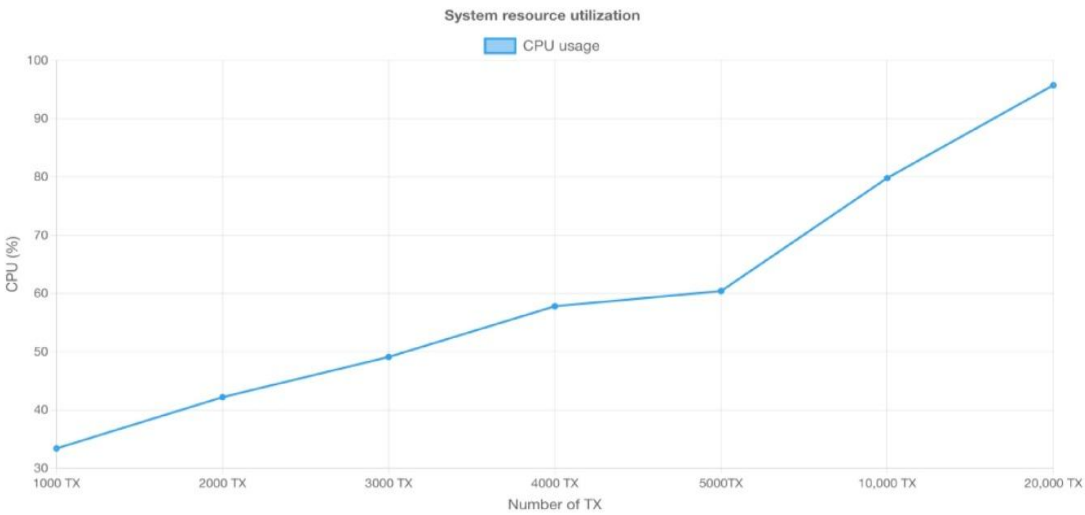

Fig 1: System resource utilization in CPU Usage graph

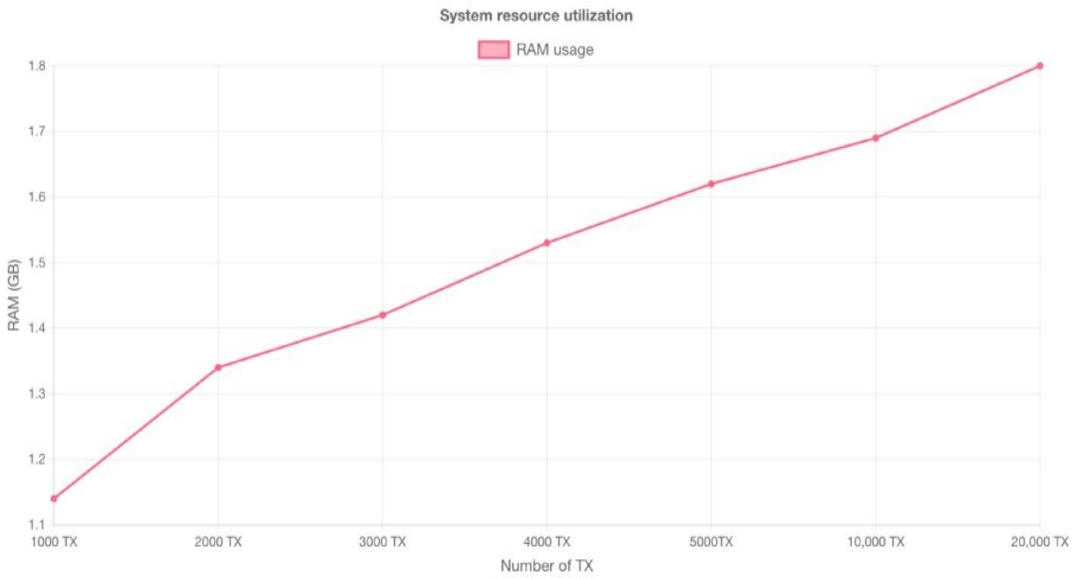

Fig 2: System resource utilization in RAM Usage graph

5. DISCUSSION

There are several privacy and security breach causes in a non-blockchain system that do not apply to the blockchain environment. Organizations accumulate a tremendous amount of data consisting of personal and confidential information for analyzing and divulging novelties in their businesses without worrying about the privacy and security effects. In addition, the data collection, storage, processing, and sharing is outsourced to third parties and thus may be entrusted to cloud service providers (CSPs), depending on their security levels, and giving rise to potential security and privacy concerns. A snapshot view of the security and privacy-preserving capabilities that blockchain can provide for IoT is illustrated in Table 1.

An interesting feature of blockchain is that individual data is only accessible upon approval from the individual and it cannot be stored for future reference. The cryptographic method is utilized to store identity proof, so it is not possible to break it, thereby creating a powerful privacy protection tool. Additionally, cryptographic identity and access control systems can prove beneficial in preventing IP Spoofing attacks. In addition, the strong use case of blockchain is regarded as the secure storage with digitally signed messages and document delivery. These kinds of applications, in order to authenticate assets and a person's identity, are being developed and authenticated in shipping, finance, insurance, and particularly in autonomous IoT environments.

By leveraging blockchain's decentralized, autonomous, and trustless paradigm, several primary security and privacy issues associated with the centralized nature of cloud can be resolved. Further, even if IoT devices are positioned very near to one another, even the connections and communications between them happen via the internet and trusted third parties. Similarly, centralized models prove costly and difficult to administer when used in heterogeneous environments such as IoT, thus giving rise to the implementation and efficiency of a blockchain-based decentralized model.

The geometric growth of network size and scalability problems of IoT can be made more secure by blockchain's consensus-driven architecture. As information in blockchain is spread across multiple devices or nodes that are chain-linked with each other, an attacker can only win if they succeed in breaching over 50% of the blockchain network. It is also important to mention that although Bitcoin—the program that used blockchain form for cryptocurrency—has had a reputation among the public regarding hacking attempts, there has been no hacking of Bitcoin transactions. Instead, the intrusions took place in systems that were storing and holding Bitcoin private keys.

IoT blockchain is aggressively being researched upon by researchers by adopting the fundamentals of blockchain methodology and building blocks and integrating them across various industries, sectors, applications, and IoT infrastructure use-cases and applications. On the basis of overall concepts of IoT security and privacy views vis-à-vis the existing model and blockchain-innovated answers, one can deduce that conventional security techniques do not align with resource-restricted nature of IoT's devices and infrastructure. IoT needs to be protected from privacy and security with light, distributed, and scalable characteristics, which blockchain can potentially offer. Yet, it is not light in nature, but computationally costly, therefore in need of adaptations and optimizations for the IoT privacy and security needs.

The individual roles within transactions are held accountable by blockchain and hence are not subject to disputes. Because blockchain is indestructible and cannot be altered or erased, it eliminates the potential for devices to connect with a network through masquerading or signature forging. In particular, device identity can be dealt with in much the same manner as the model of cryptocurrency to grant ownership in bitcoins; it can be utilized for IoT devices and users to a greater degree. The decision between permissioned and permissionless blockchain is based on different types and quantities of authorizing participants with different credentials, as well as the type and significance of assets being transferred or exchanged.

It is clear from the above discussion that blockchain can be a nightmare for cybercriminals, data manipulators, and others who abuse personal data and devices. Even as blockchain is being predicted to find solutions to various security and privacy issues, its most promising future is the owner of personal information controlling it. For instance, once a digitally signed copy of a document such as a driving license, the ownership documents for a vehicle, house, piece of land, and other such assets, etc., is put into blockchain by a government authority, it will be unerasable and unreadable. Also, it must be considered that blockchain is still in its nascent stage. The volume of applications using distributed ledger technology (DLT) and blockchain is changing to properly assess the superiority of this technology over existing systems in protection against different cyberattacks, threats, and risks.

Nevertheless, the architecture and design of blockchain infrastructure are promising for the improvement of IoT smart applications.

6. FUTURE SCOPE

The future potential of blockchain tokenization is vast, especially as it increasingly connects with emerging technologies such as artificial intelligence and decentralized finance (DeFi). By applying mathematical models similar to those found in deep learning—where transformations and activations influence outcomes—tokenization can evolve into more programmable and dynamic financial instruments. As assets are separated and represented as tokens, smart contracts permit the automation of ownership logic, distribution, and governance, boosting transparency and efficiency. This innovation makes the path for wider adoption across industries like real estate, intellectual property, and supply chain management. As tokenization continues to grow, it has the potential to transform our notion of value, ownership, and exchange within a digital economy.

The suggested Massive IoT architecture, although displaying promise performance and scalability, opens various options for further refinement. A major area for improvement is the introduction of edge computing, which can minimize latency and bandwidth use by running computations closer to the data source. Additionally, the employment of AI-optimized algorithms could allow for real-time adjustments to system features such as payload size, transaction rate, and resource allocation, thereby boosting efficiency and dependability Future advances may also include compatibility with other blockchain systems to facilitate cross-chain data transfer and enhance system flexibility. Regarding security, implementing zero-knowledge proofs or multi-signature methods would increase privacy and access control in multi-user contexts. Lastly, scaling the architecture for production in multi-region deployments and analyzing performance under various network situations would provide vital insights into its practical usability and robustness.

7. CONCLUSION

Given the delicacy linked to the performance of smart devices and the data they gather, create, and process, it is important that IoT solutions address possible security and privacy concerns. It should be noted that IoT security and privacy concepts develop in line with changing trends, technology development, and the exact requirements of different industries or applications. Blockchain technology not only enables monetary transactions but also safeguards the transfer of data and resource allocation between devices, making it a competent service for IoT purposes. The networked world can help implement Blockchain technology as an extra layer, which is able to assist an increasing range of devices, such as wearables, sensors, smartphones, tablets, laptops, homes, cars, and smart cities, through its upgrading features. This research brings forward the drawbacks of IoT and useful features of Blockchain that will support the desired privacy and security within IoT infrastructures and applications. It has been reported that there are several optimistic possibilities in using Blockchain for future IoT. The challenges are also there, most importantly related to consensus models as well as to the computational load of verifying transactions. In spite of such challenges, Blockchain technology is only in its budding stages, and such issues will be solved soon, allowing numerous opportunities to shape concepts into material realities.

8. REFERENCES

[1] M. H. Rehman, M. A. Khan, T. Hussain, Y. Nam, and S. Rho, “Integrating blockchain and IoT for secure and scalable networks: A comprehensive survey,” *IEEE Internet of Things Journal*, vol. 8, no. 5, Mar. 2021.

[2] Novo, Oscar. "Blockchain meets IoT: An architecture for scalable access management in IoT." *IEEE internet of things journal* 5, no. 2 (2018): 1184-1195.

[3] Dorri, Ali, Salil S. Kanhere, Raja Jurdak, and Praveen Gauravaram. "Blockchain for IoT security and privacy: The case study of a smart home." In *2017 IEEE international conference on pervasive computing and communications workshops (PerCom workshops)*, pp. 618-623. IEEE, 2017.

[4] M. Ali, M. Asim, S. Islam, and N. Javaid, “Blockchain-based authentication mechanism for vehicular IoT networks,” *Sensors*, vol. 20, no. 12, Jun. 2020.

[5] X. Xu, I. Weber, and M. Staples, “Architecture for blockchain-based industrial Internet of Things,” in *Proc. IEEE Int. Conf. on Software Architecture (ICSA)*, Seattle, WA, USA, Apr. 2018.

[6] V. Sharma, I. You, and R. Kumar, “Is blockchain the answer to edge-based IoT security and privacy?” *IEEE Communications Magazine*, vol. 58, no. 9, Sep. 2020.

[7] Conoscenti, Marco, Antonio Vetro, and Juan Carlos De Martin. "Blockchain for the Internet of Things: A systematic literature review." In *2016 IEEE/ACS 13th International Conference of Computer Systems and Applications (AICCSA)*, pp. 1-6. IEEE, 2016.